OpenShift StatefulSet Guidelines and Best Practices

StatefulSet

Summary

The following describes guidelines and best practices, specific to the Kubernetes "StatefulSet" resource. This is a supplement to the OpenShift General Workload Guidelines and Best Practices.

Reference Documentation

- OpenShift General Workload Guidelines and Best Practices
- StatefulSet [apps/v1] - Workloads APIs | API reference | OpenShift Container Platform 4.16

StatefulSet Best Practices

- Declare replicas greater than 0
- Declare a minimum of 3 Replicas in UAT, Perf and Production Environments
- Declare Anti-affinity rules
- Only use a StatefulSet when an application requires one or more of the following. Otherwise, use a Deployment.
    - Stable, unique network identifiers
    - Stable, persistent storage
    - Ordered, graceful deployment and scaling
    - Ordered, automated rolling updates
- Use unique and meaningful names for your StatefulSets and their Pods
- Use headless services for StatefulSets
- Define a Pod Disruption Budget (PDB)
- Define Vertical Pod Autoscaling (VPA) in Inform mode
- Use Affinity rules to minimize cross-node traffic between dependent services
- Use a **volumeClaimTemplate** over a RWX PVC
- If exposing as a webservice, label ports within the Container spec for the pod(s) (container.ports.name) and reference the port names from the Service object

StatefulSet Anti-Patterns

- Single Replica
- Use of Horizontal Pod Autoscaling (HPA)

Development Best Practices

- Test with a minimum of 2 replicas in lower environments